School of Computing Science & Engineering

Department of Computer Science & Engineering

Academic Year: 2025-2026

Deep Learning and Its Application

Assignment-1

PROJECT SYNOPSIS

On

Non-Invasive Anemia Detection on Ocular Images

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Project Guide: Dr. Rashmi Rathi Upadhyay

1. Introduction and Background

- Anemia is a major global health issue affecting over 1.8 billion people worldwide, particularly women and children in developing regions. It occurs due to insufficient hemoglobin levels in the blood, leading to fatigue, weakness, and severe medical complications.
- The conventional diagnostic method involves invasive blood testing, which requires laboratory infrastructure, skilled personnel, and sterile conditions. This makes large-scale screening difficult in rural and low-resource settings.
- Recent advancements in Deep Learning and Computer Vision have enabled non-invasive diagnostic systems using medical imaging. Studies indicate a strong correlation between hemoglobin levels and pallor in the palpebral conjunctiva of the eye.
- This project leverages Deep Learning-based Convolutional Neural Networks (CNNs) and ensemble learning techniques to develop a reliable non-invasive anemia detection system using ocular images.

2. Problem Statement

Current anemia detection methods rely on invasive blood tests, which are expensive, time-consuming, and inaccessible in rural regions.

Existing deep learning models:

- Show bias due to class imbalance.
- Struggle with low recall for the anemic class.
- Lack robust generalization.

There is a need for a cost-effective, non-invasive, and accurate deep learning-based screening system that improves recall for detecting anemic patients.

3. Objectives

Primary Objective:

To develop a reliable non-invasive anemia detection system using ensemble deep learning techniques.

Specific Objectives:

1. To preprocess ocular images using normalization and CLAHE.
2. To implement transfer learning using VGG16, DenseNet121, and InceptionV3.
3. To address class imbalance using ensemble learning.
4. To maximize recall for the anemic class.
5. To evaluate performance using accuracy, precision, recall, F1-score, and MSE.

4. Literature Review

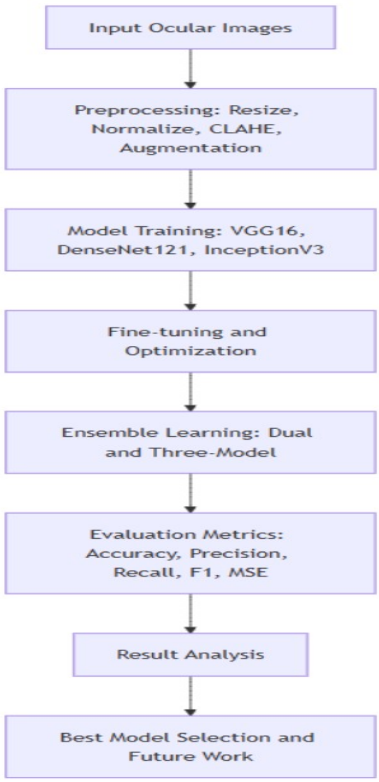
- Sehar (2025) – Used deep learning on conjunctiva images for anemia detection.
- Mahmud et al. (2023) – Applied transfer learning on lip mucosa images.
- Dimauro et al. (2023) – Used sclera-based feature extraction for anemia detection.
- Hettiarachchi et al. (2023) – Developed lightweight model for smartphone-based anemia screening.
- Liu & Li (2024) – Applied Vision Transformers (ViTs) for ocular anemia detection.

Research Gap:

- Most models suffer from class imbalance bias.
- Standalone models show low recall for anemic cases.
- Limited use of ensemble learning for improved diagnostic stability.

5. Proposed Methodology

- System Architecture



- Dataset Description

  1. Ocular images focusing on conjunctiva and sclera.
  2. Class imbalance ratio: Approximately 3:1 (Non-anemic : Anemic).
  3. Images resized to 224×224 pixels.

- Data Preprocessing Steps

  1. Image resizing (224×224)
  2. Min-Max normalization
  3. CLAHE (Contrast Limited Adaptive Histogram Equalization)
  4. Data augmentation (rotation, zoom, horizontal flip)

- Model Architecture

1. VGG16 (baseline feature extractor)
2. DenseNet121 (feature reuse via dense connections)
3. InceptionV3 (multi-scale convolution)
4. Soft Voting Ensemble (probability averaging)
5. Final probability:

   $$P_final = (P1 + P2 + P3) / 3$$

- Training Strategy

1. Transfer Learning with ImageNet weights
2. Optimizer: Adam
3. Learning Rate: 1×10^{-4}
4. Batch Size: 32
5. Epochs: 50
6. Early Stopping
7. 5-fold Cross Validation
8. Loss Function: Binary Cross-Entropy

- Evaluation Metrics

1. Accuracy
2. Precision
3. Recall (Focus on anemic class)
4. F1-score
5. Mean Square Error (MSE)

6. Tools and Technologies

Programming Language:

- Python 3.10

Frameworks:

- TensorFlow
- Keras

Hardware Requirements:

- NVIDIA RTX 3060 GPU
- 12GB VRAM
- 16GB RAM minimum

7. Expected Outcomes

- Expected Accuracy: ~80% to 90%
- Anemic Recall: ~0.70 to 0.80
- Reduced bias toward majority class.
- Reliable non-invasive screening tool.
- Potential deployment in rural healthcare and mobile-based screening systems.
- Scope for research publication in biomedical AI domain.

8. Work Plan and Timeline

Phase	Activity	Duration
Phase 1	Literature Review	2 Weeks
Phase 2	Dataset Preparation & Preprocessing	2 Weeks
Phase 3	Model Development	3 Weeks
Phase 4	Ensemble Integration	2 Weeks
Phase 5	Testing & Validation	2 Weeks
Phase 6	Documentation & Report	1 Week

9. References

- [1] N. Sehar, "Detection of Anemia Using Conjunctiva Images with Deep Learning Model," Health Informatics Research, 2025.
- [2] S. Mahmud et al., "Non-invasive detection of anemia from lip mucosa images using transfer learning," PLoS ONE, 2023.
- [3] G. Dimauro et al., "Anemia Detection Based on Sclera and Color Estimation," Biomedical Signal Processing and Control, 2023.
- [4] H. Hettiarachchi et al., "Lightweight Deep Learning Model for Real-Time Anemia Screening," IEEE JBHI, 2023.
- [5] W. Liu and Z. Li, "Transformer for Non-invasive Anemia Detection," Computers in Biology and Medicine, 2024.